Project: Small Office Network Design (25-User Office)

A guided project with real deliverables.

Scenario

You're the network engineer for **BrightPath Consulting**, a 25-employee office that is moving into a new single-floor space. They need a secure, reliable network that can grow to about 40 users over three years.

Client requirements

- 25 staff PCs and laptops, 3 network printers, 1 file server (or NAS), 6 IP phones
- Guest Wi-Fi that is isolated from the internal network
- Departments: Management (4), Sales (8), Accounts (6), IT/Support (3), Reception (4)
- Accounts and Management traffic must be separated from other departments
- Single ISP link, with a basic plan for failover
- Budget: about ₹3-4 lakh for hardware (or $4,000-5,000)

Phase 1: Requirements and Planning (Day 1-2)

Do: Interview-style requirements gathering, a site survey, and a bandwidth estimate.

Deliverables

1. Requirements document (users, devices, services, growth, budget)
2. Floor plan sketch showing rack location, cable runs, and Wi-Fi access point placement
3. Bandwidth estimate (e.g., 25 users × 5 Mbps average = about 125 Mbps, with a justified ISP plan)

Phase 2: Logical Design (Day 3-4)

Deliverables

1. **Topology diagram** (draw.io, Visio, or Packet Tracer)
2. **VLAN plan**

VLAN	Name	Subnet	Purpose
10	Management	192.168.10.0/26	Management/executives

VLAN	Name	Subnet	Purpose
20	Sales	192.168.20.0/26	Sales team
30	Accounts	192.168.30.0/27	Finance, restricted
40	IT	192.168.40.0/28	IT admin
50	Voice	192.168.50.0/27	IP phones
60	Servers/Printers	192.168.60.0/27	NAS, printers
70	Guest	192.168.70.0/26	Isolated Wi-Fi
99	Network Mgmt	192.168.99.0/28	Switch/AP management

3. **IP addressing scheme** with reserved ranges for gateways, static devices, and DHCP pools
4. **Security policy table** (who can talk to whom), for example: Guest → Internet only; Sales → Servers (file shares only); Accounts → Servers + restricted share; IT → everything

Phase 3: Physical Design and Bill of Materials (Day 5)

Deliverables

1. Equipment list with justification:
   - 1× router/firewall (e.g., Fortinet 60F, MikroTik, pfSense box)
   - 1× 24-port PoE managed switch + 1× 8-port access switch (for growth)
   - 2-3× Wi-Fi 6 access points
   - Patch panel, 12U wall rack, UPS
   - Cat6 cabling plan (drops per desk, meeting room, printers)
2. **Costed BOM** (spreadsheet with unit price, quantity, total, and 15% contingency)
3. Cable schedule (port-to-port labeling scheme)

Phase 4: Build and Simulate (Day 6-8)

Build the design in **Cisco Packet Tracer**, **GNS3**, or **EVE-NG**.

Configure

- VLANs and trunk ports
- Inter-VLAN routing (router-on-a-stick or L3 switch SVIs)
- DHCP pools per VLAN

- ACLs enforcing your security policy
- NAT/PAT for internet access
- Port security and disabling unused ports
- Wireless SSIDs (Staff and Guest) mapped to VLANs

Deliverables

1. Working simulation file (.pkt / .gns3project)
2. Full device configs (text files)
3. Test plan with results, e.g.:

Test	Expected	Result
Sales PC pings Accounts PC	Blocked	
Guest reaches internal server	Blocked	
All VLANs reach internet	Success	
Phone gets DHCP in VLAN 50	Success	

Phase 5: Security, Monitoring, and Resilience (Day 9)

Deliverables

1. Firewall rule set and hardening checklist (SSH only, no default passwords, SNMPv3, NTP, logging)
2. Backup plan: config backups, NAS backup (3-2-1 rule)
3. Monitoring setup: Zabbix, PRTG free tier, or LibreNMS, with a screenshot of the dashboard
4. Failover plan: secondary 4G/5G link or a second ISP, with a described switchover method

Phase 6: Documentation and Handover (Day 10)

Final package

1. Network design document (PDF) covering everything above
2. As-built diagrams (logical and physical)
3. IP/VLAN spreadsheet
4. Admin runbook: how to add a user, reset a switch, replace an AP, and troubleshoot common issues

5. 10-minute presentation to a "non-technical client"

Grading Rubric (100 pts)

Area	Points
Requirements and planning	10
Logical design (VLANs, IPs)	20
Physical design and BOM	15
Working simulation and configs	25
Security and monitoring	15
Documentation and presentation	15

Stretch Goals

- Add a site-to-site VPN to a branch office
- Implement 802.1X/RADIUS authentication for Wi-Fi
- Add a redundant switch with STP/LACP link aggregation
- Build a small Ansible/Python script to back up configs automatically
- Plan an IPv6 rollout

Tools

Packet Tracer (free), draw.io, Excel/Sheets, Wireshark, Zabbix/LibreNMS, and a PDF or Word editor for the final document.